

Custom Silicon Programs are Inherently Difficult to Operate

Industry context

As AI models become more compute-intensive, high-compute companies are increasingly investing in custom silicon tailored to their training and inference workloads. Delivering these chips requires close coordination with semiconductor partners across design, testing, validation, and manufacturing.

Target persona

Semiconductor companies running custom-silicon programs for high-compute customers, such as Broadcom, Micron, and Intel.

Business problem

Requirement changes and yield excursions ripple quickly across validation, supply, schedule, and customer commitments. Teams must reconcile fragmented evidence across multiple systems to understand impact, protect yield, and decide how to respond.

Current challenges

Requirements evolve throughout the program. As model sizes, context windows, throughput targets, and other requirements change, customers ask the program to support new or updated workloads.

Impact assessment is fragmented. The evidence needed to understand a change is spread across engineering, validation, manufacturing, planning, and commercial systems, slowing confident decisions.

Execution is fragmented across systems. Once a decision is made, the required actions span multiple systems making coordination, follow-through, and verification difficult.

Program exposure grows. Slow impact assessment leads to missed milestones, unnecessary validation, inventory exposure, unreliable commitments, and revenue at risk.

\$203.6B

Total market for custom semiconductor ASIC market by 2030

\$10M+

1% drop in yield percentage causes tens of millions of dollars in revenue loss

\$2M

Per day of market opportunity at risk if deadlines are missed

⚠️ As-is: manual, time-consuming, and error-prone

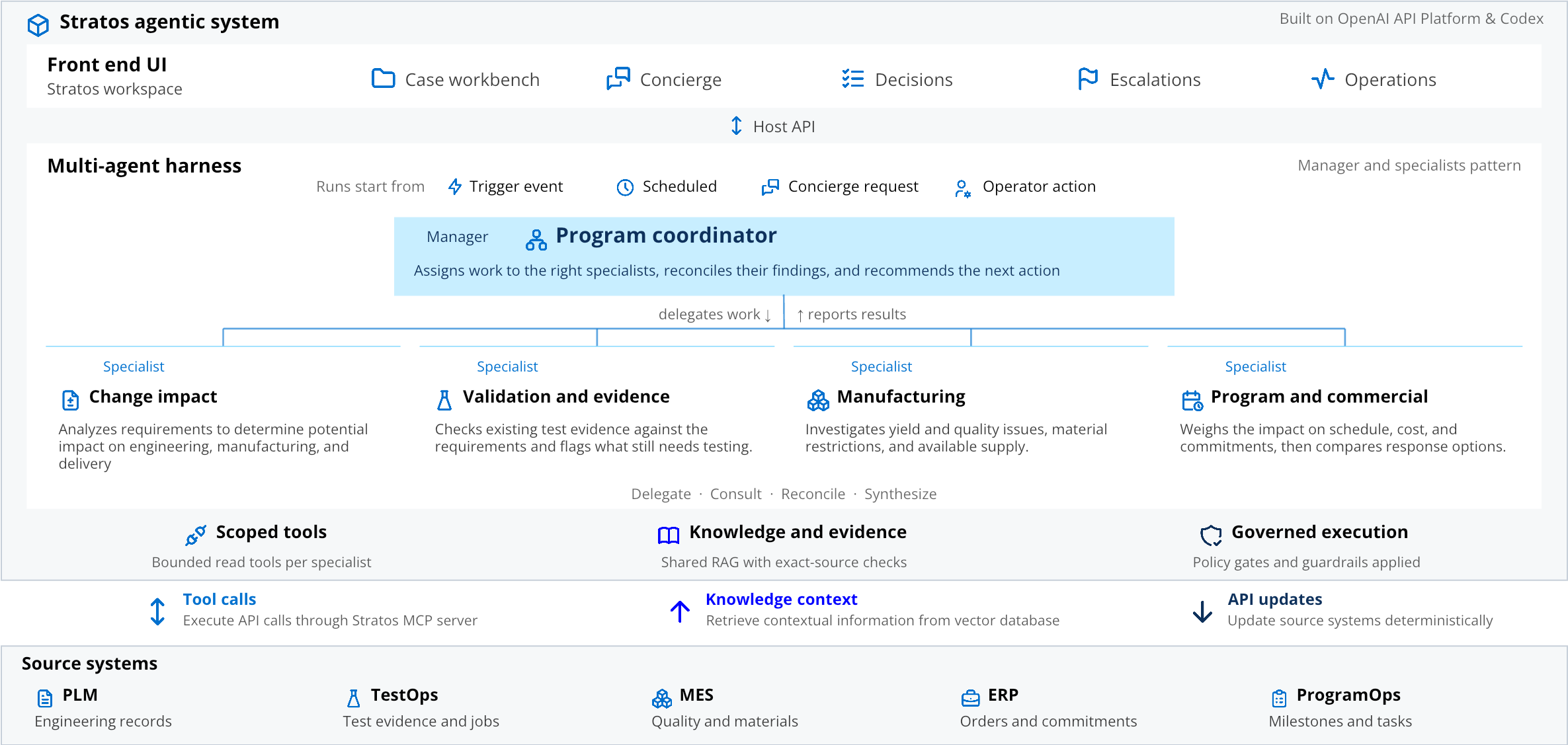
👤 **Program operator** chases every stage by hand: emails, portals, and spreadsheets to learn what changed, what is ready, and who must approve next



The Solution Architecture

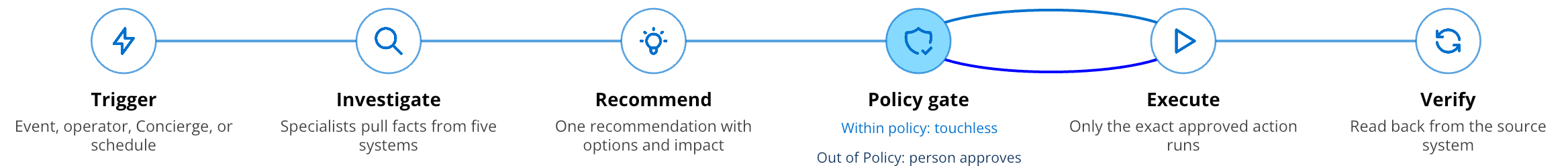
OpenAI

The OpenAI demo solution built for our imaginary customer: Stratos Silicon



Demo Use Cases: Turning Requirement Changes into Verified Actions Autonomously

To-be flow: Agents autonomously process in the background and escalate to program approver when needed



Demo use cases:

Standard requirement change

CR-019 **Autonomous**

A customer requests an update within an approved test profile. Agents reuse the matching evidence and route the validation package.

Material requirement change

CR-017 **Engineering approval**

A customer requests a much broader workload than existing tests cover. Agents find the gaps, analyzed next best actions, routed to engineering, and updated plan to ensure on-time completion

Yield and quality recovery

QE-004 **Program Owner approval**

A yield exception alert raises questions about usable supply and timing. Agents size the impact and recommends updated supply chain actions downstream

Business value drivers*:

Yield protected

\$20M

of saleable output protected per year, for every yield point recovered

Time to decision

5 days to 1

from detection to an actionable, evidence-backed decision

Commitment reliability

\$25M

of shipment value per year moved out of commitment risk

Working capital

\$4.1 M

released from 10 fewer days of inventory waiting on decisions

Demo data assumptions*: 7,500 wafers and 45 dies at \$6K each; a \$500M annual shipment run-rate; late commitments falling from 10% to 5%; \$150M average inventory. To be validated in a pilot. Exposure removed from risk is not realized savings.

Eight enterprise design decisions behind the control plane

Category	Enterprise design decision
1  Agent Architecture & Orchestration	A central coordinator delegates to domain specialists that run in parallel. Specialization and concurrency without an uncontrolled swarm.
2  Grounding & Enterprise Knowledge	Live facts come from APIs and MCP; institutional knowledge comes from RAG with specialist retrieval profiles and exact-source verification. Stale documents never override current state.
3  Tooling & System Integration	Agents reach PLM, TestOps, MES, ERP, and ProgramOps only through typed tools and controlled APIs. Source systems stay authoritative; the control plane orchestrates across them.
4  Governance, Permissions & Human Authority	Deterministic policy gates decide what runs touchless and what needs approval. Engineering, quality, allocation, and customer commitments stay human-owned, backed by least-privilege tools and persona-aware RBAC.
5  State, Execution Safety & Reliability	Approval, execution, and verified state are separate steps. Versioned proposals, stale-state checks, idempotency, and source readback make autonomous and concurrent execution safe.
6  Evals, Observability & Auditability	Behavioral evals test workflow correctness, retrieval, authority boundaries, and calculations. Traces link agents, tools, evidence, decisions, actions, and outcomes for reliability and audit.
7  Performance, Cost & Scalability	Models handle ambiguous reasoning; deterministic code handles calculations, policy, and routine operations. Isolated, bounded runs scale toward queues and worker pools without wasting inference.
8  Product UX & Operating Model	Manual, conversational, and event-driven starts share one governed runtime. Persona-tailored screens show autonomy, handoffs, evidence, and business state, not raw agent mechanics.

Production hardens the same architecture. Autonomy expands only as measured results allow.

Connect and secure

Identity, real systems, data

Authentication, SSO & RBAC

Integrate with enterprise identity providers for user authentication/SSO, map roles and groups to RBAC, and enforce least-privilege access to agents, tools, data, and actions.

Real Enterprise Integrations

Connect production PLM, MES, ERP, TestOps, planning, and internal systems while keeping those systems authoritative.

Security & Data Governance

Add secrets management, encryption, data isolation, retention policies, audit logging, and prompt/tool abuse protections.

Run reliably at scale

Routing, runtime, safe execution

Dynamic Model Routing

Route each task to the right model based on required quality, complexity, latency, and cost, with fallback/escalation when needed.

Durable Runtime & Horizontal Scale

Add durable queues/workflow orchestration, stateless workers, checkpoints, retries, and support for many concurrent cases.

Execution Safety & Resilience

Preserve idempotency, version checks, stale-state protection, partial-failure handling, retries, and source readback verification.

Improve and expand

Monitoring, evals, earned autonomy

Observability & Evaluation-Driven Improvement

Use OpenAI Platform telemetry for traces, usage, latency, and cost, then add Stratos workflow outcomes, RAG/tool quality, and regression evals to continuously improve reliability.

Measured Expansion of Autonomy

Use those technical and business signals to decide when a workflow can safely move from assistive to touchless execution, based on predefined safety, quality, and business-value thresholds.